

Sequences & Series



Arithmetic sequences

- 1 For the arithmetic sequence with $a_1 = 9$ and common difference $d = 6$, find a_{10} .
 - 2 For the arithmetic sequence with $a_1 = 15$ and $d = -4$, find a_{11} .
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Arithmetic series

- 3 Find S_{10} for the arithmetic sequence with $a_1 = 9$, $d = 6$ (using $a_{10} = 63$ from above).
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Geometric sequences

- 4 For the geometric sequence with $a_1 = 6$ and common ratio $r = 2$, find a_8 .
 - 5 A geometric sequence has $a_1 = 2$ and common ratio $r = 5$. Find the smallest n for which $a_n = 1250$.
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Geometric series (finite)

- 6 Find S_8 for the geometric sequence with $a_1 = 6$, $r = 2$ (using $a_8 = 768$ from above).
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Infinite geometric series

- 7 Find the sum of the infinite geometric series $36 + 12 + 4 + \dots$.
- 8 Find the sum of the infinite geometric series $12 - 4 + \frac{4}{3} - \dots$.